ECBM E4040 COLUMBIA UNIVERSITY FALL 2023

# Neural Networks and Deep Learning

Course syllabus

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| Instructor | Mehmet Kerem Turkcan | Time | Fridays, 10:10 AM to 12:40 PM |
| Email | mkt2126@columbia.edu | Location | 501 Northwest Corner Building |

## Course description

The course covers the theoretical underpinnings and practical aspects of neural networks and deep learning. Students learn about and implement a range of deep learning architectures, including convolutional and recurrent neural networks. The focus of the course is on applications and projects.

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| Courseworks | https://courseworks2.columbia.edu/courses/180017 |
| Ed Discussion forum | https://edstem.org/us/courses/46398/discussion/ |
| Course directory | https://doc.sis.columbia.edu/#subj/ECBM/E4040-20233-001/ |
| Office hours | After class, by appointment, or by posting questions on the Ed Discussion forum. |
| Teaching assistants | Chengbo Zang, Aishwarya Patange, Sanjeev Narasimhan, and William Ho. |
| TA and CA office hours | Announced on Courseworks. |

## References and prerequisites

The main resource for the course is *Deep Learning* by Ian Goodfellow, Yoshua Bengio, and Aaron Courville (MIT Press, 2016), available at deeplearningbook.org. Students should have taken machine learning, previously or in parallel with this course, and need a background in probability and statistics, and in linear algebra.

# Grading

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| Assignments, 40% | Four graded assignments. Each student completes their own coding tasks and questions, using Python, TensorFlow, Jupyter notebooks, and code management tools. |
| Exam, 25% (week 11) | An in-class exam covering deep learning theory, architecture, and design, as well as the prerequisite topics in linear algebra, probability and statistics, calculus, and basic machine learning. |
| Final project, 35% | One project based on contemporary papers or original ideas, in groups of no more than three, documented in code and a report. |

**Late assignments: slip days**

Each student has four late days without penalty, to divide among the assignments in any way. For example, assignment 2 may be four days late, in which case no other assignment may be late; or assignment 1 may be one day late and assignment 2 two days late, leaving one late day for later assignments. Late days cannot be divided into hours. Requests for additional extensions will not be granted: once the four days are used, late assignments receive no credit. The late policy does not apply to project submissions.

# Course outline

| Week | **Topics** |
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| 1 | Introduction to the course and to deep learning; computing resources for deep learning |
| 2 | Introduction to the course (continued); review of machine learning |
| 3 | Deep feedforward networks; introduction to backpropagation |
| 4 | Backpropagation; t-SNE; the universal approximation theorem |
| 5 | Optimization for deep learning |
| 6 | Convolutional neural networks |
| 7 | Regularization |
| 8 | Deep learning in practice |
| 9 | Modern CNN architectures |
| 10 | Recurrent neural networks; midterm preparation |
| 11 | **Midterm exam** |
| 12 | Autoencoders |
| 13 | GANs, variational autoencoders, and trends in deep learning |
| 14 | GANs, variational autoencoders, and trends in deep learning (continued) |

# Important dates

| Date | **Due** |
|---|---|
| Sep 22 | Assignment 0 |
| Oct 17 | Assignment 1 |
| Nov 7 | Assignment 2 |
| Nov 17 | **Midterm exam** |
| Nov 28 | Assignment 3 |
| Dec 17 | **Final project** |